arises the **pulmonary trunk** which bifurcates to form right and left pulmonary arteries. The right and left pulmonary arteries supply **deoxygenated blood** to the lungs of the respective side. The left ventricle is longer and narrower than the right ventricle. The walls are about three times thicker than the right ventricle. The **left ventricle** gives rise to **aorta**. The **oxygenated blood** is supplied by the aorta to various organs of the body. The **coronary arteries** supply blood to the heart.

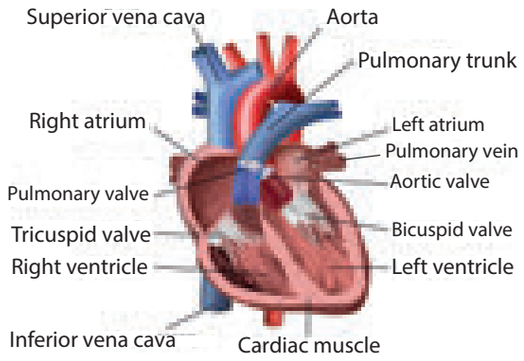



**Figure. 14.12** Internal structure of human heart

**Valves:** The valves are the muscular flaps that regulate the flow of blood in a single direction and prevent back flow of blood. The heart contains three types of valves.

**Right atrioventricular valve:** It is located between the right auricle and right ventricle. It has three thin triangular leaf like flaps and therefore called **tricuspid valve.** The apices of the flaps are held in position by **chordae tendinae** arising from the muscular projection of the ventricle wall known as **papillary muscles.**

**Left atrioventricular valve:** It is located between the left auricle and left ventricle. It has two cusps and therefore called **bicuspid** or **mitral valve.**

## More to Know

**Heart chambers in vertebrate animals**
**Two chambered:** Fishes
**Three chambered:** Amphibians
**Incomplete four chambered:** Reptiles
**Four chambered:** Aves, Mammals and Crocodiles (Reptile)

**Semilunar valves:** The major arteries (pulmonary artery and aorta) which leave the heart have semilunar valves which prevent backward flow of blood into the ventricles. They are the pulmonary and aortic semilunar valves.

## 14.15.1 Types of Blood Circulation

The blood circulates in our body as oxygenated and deoxygenated blood. The types of circulation are:

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i **Systemic circulation**: Circulation of oxygenated blood from the left ventricle of the heart to various organs of the body and return of deoxygenated blood to the right atrium. Aorta carries oxygenated blood to all the organs of the body.

ii **Pulmonary circulation:** The path of pulmonary circulation starts in the right ventricle. Pulmonary artery arises from the right ventricle and reaches the lungs with deoxygenated blood. Pulmonary veins collect the oxygenated blood from the lungs and supplies it to the left atrium of the heart.

iii **Coronary circulation:** The supply of blood to the heart muscles (cardiac muscles) is called as **coronary circulation.** Cardiac muscles receive oxygenated blood from **coronary arteries** that originate from the **aortic arch.** Deoxygenated blood from the cardiac muscles drains into the right atrium by the **coronary sinuses.**

When the blood circulates twice through the heart in one complete cycle it is called **double circulation.** In double circulation the oxygenated blood do not mix with the deoxygenated blood.

However, in some animals the oxygenated and deoxygenated blood are mixed and pass through the heart only once. This type of circulation is called **single circulation**. e.g., fishes, amphibians and certain reptiles.



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